

Kiew Yao Hong

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SUMMARY & SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, NestJS, Node.js, Express, FastAPI

Frontend & Tools: React, Astro, HTML, CSS, Tailwind, Docker, Pivotal Cloud Foundry (PCF), AWS, Git

EXPERIENCE

DBS Bank

Jul 2024 – Present

Software Engineer – Associate

- Migrated Cards and Payments services, focused on vendor integration (VisionPLUS, PowerCARD) between core banking systems and transitioning services to a cloud-native platform, increasing in-house ownership and reducing operational costs.
- Designed API replay system leveraging ETL pipelines to extract and transform API network traffic for persistent storage, enabling replay-based debugging and reducing turnaround time for diagnosing integration issues.
- Integrated card processing systems with payment schemes (Visa, Mastercard), providing transaction processing capabilities to downstream banking services.
- Lead end-to-end change and risk management for the team, overseeing 10+ production changes monthly, coordinating technical review, risk assessment and cross-regional management approval.
- Implemented transaction security component for fraud detection API triggered on transaction events to identify and flag unauthorized retailer activity

NCS Group

Nov 2022 – Jun 2024

Software Engineer

- Modernized legacy systems to cloud-native architecture on AWS, improving reliability, maintainability, and regulatory compliance under Whole of Government (WoG) initiative.
- Delivered full-stack web application using React and Spring Boot, refactoring API endpoints, optimizing database queries, and leveraging AWS CloudWatch to reduce incident response time by 50%.
- Led the redesign of React components into reusable, modular patterns, consolidating over 20 static pages into a configurable dynamic module improving Google Lighthouse performance scores from 80 to 90+ while reducing code duplication and improving developer maintainability.
- Proposed and implemented system enhancements aligned with evolving client needs, improving solution adaptability and contributing to long-term revenue growth.

PROJECTS AND MISCELLANEOUS

Expense Application | React, Java, Docker, PostgreSQL

- Started as a teaching project for an online developer community and evolved into a fully functional finance platform through iterative development.
- Implemented transaction workflows with consistent frontend–backend state synchronization, experimenting with asynchronous event processing to explore system decoupling.

Pathfinding Visualiser | Python, Pygame

- Built interactive grid-based graph visualizer supporting dynamic wall placement and start/end node configuration.
- Implemented Dijkstra visualization to animate node exploration and shortest distance

Comment Summary | Python, FastAPI, React, TypeScript

- Developed a web tool assisting viewers to quickly understand the sentiment and discussion trends of any YouTube video without manually reading through hundreds of comments.
- Implemented summarisation via Groq's model to generate concise insights about audience sentiment and discussion themes.

EDUCATION

University of Wollongong

Computer Science, Cybersecurity

Singapore Institute of Management

Sep 2020 – Mar 2022

Temasek Polytechnic

Diploma in Information Technology

Apr 2015 – May 2018